

Generative AI and Agentic AI

Documentation Report

AI Internship Preparation Notes with Practical Examples, Comparisons, Technologies, Use Cases, and Interview Answers

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Focus Area: Generative AI, Agentic AI, RAG, Tools, Memory, Planning, and AI Frameworks

Core idea: Generative AI creates content. Agentic AI works toward goals using planning, memory, tools, and actions.

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1. Learning Objective and Scope

This documentation explains Generative AI and Agentic AI from beginner level to interview-ready understanding. The goal is to understand what these concepts mean, how they work, what technologies are used, where they are applied, and how to explain them confidently in academic or internship interviews.

Learning focus: understand concepts clearly first, then connect them with practical tools, projects, and real-world use cases.

- Understand what Generative AI is and what it can create.
- Understand what Agentic AI is and how it differs from normal chatbots.
- Learn basic technologies such as LLMs, prompts, tokens, embeddings, vector databases, and RAG.
- Learn Agentic AI concepts such as tools, memory, planning, actions, and workflows.
- Understand frameworks such as LangChain, LangGraph, CrewAI, LlamaIndex, and OpenAI Agents SDK at a basic level.
- Prepare interview-ready answers and examples.

2. Introduction to Generative AI

Generative AI is a type of artificial intelligence that can create new content based on user prompts. This content may include text, code, images, audio, video, summaries, reports, documents, and presentations. It is called generative because it generates new output instead of only classifying or predicting labels.

Simple definition: Generative AI is AI that generates new content from a given prompt.

Content Type	Example Output
Text	Emails, blogs, summaries, reports, documentation
Code	Python code, JavaScript code, FastAPI routes, SQL queries
Image	Posters, logos, product mockups, illustrations
Audio	Text-to-speech, voice generation, dubbing
Video	AI-generated clips, video summaries, explainers
Data	Sample JSON data, test cases, synthetic datasets

Example: If the prompt is 'Write a professional email for internship request', a Generative AI system can create a complete email draft with subject, greeting, body, and closing.

3. Basic Working of Generative AI

Generative AI models are trained on large datasets. A text-based model learns patterns in language and generates responses by predicting suitable next tokens based on the context. In simple words, the model reads the input prompt, understands the context, and generates an output that matches the user request.

Step	What Happens
1. User Prompt	The user gives an instruction or question to the AI model.
2. Processing	The model converts the text into tokens and analyzes the context.
3. Pattern Use	The model uses learned language patterns from training data.

4. Generation	The model generates new text, code, summary, or other output.
5. Response	The final answer is returned to the user or application.

Important note: Generative AI does not always know whether its output is correct. Important outputs should be verified.

4. Important Building Blocks of Generative AI

4.1 Large Language Model (LLM)

An LLM, or Large Language Model, is an AI model trained on large amounts of text data. It can understand prompts and generate human-like text, code, explanations, summaries, and answers.

- LLM is the text-generation brain of many Generative AI applications.
- It is used in chatbots, documentation tools, code assistants, summarization tools, and question-answering systems.
- A chatbot is an application; the LLM is the model behind it.

Interview line: An LLM is the core AI model, while a chatbot is an application that uses the LLM to interact with users.

4.2 Prompt

A prompt is the instruction or input given to an AI model. The quality of the prompt directly affects the quality of the output. A clear prompt gives task, context, format, tone, and constraints.

Prompt Element	Purpose	Example
Role	Defines model behavior	Act as an AI internship mentor
Task	Tells what to do	Explain RAG
Context	Adds background	I am a beginner preparing for interview
Format	Controls output	Give bullets and one example
Constraint	Sets limits	Use simple English

Good prompt example:

Act as an AI teacher. Explain Generative AI in simple English.

Include one real-life example, one short definition, and one interview-ready answer.

4.3 Token

A token is a small unit of text processed by a language model. It can be a word, part of a word, or punctuation. LLMs process both input prompts and output responses in tokens.

- Input tokens come from the prompt and context.
- Output tokens are generated by the model.
- Token count affects response length, context window, and API cost.

- A context window is the total amount of text the model can handle at one time.

4.4 Embedding

Embedding means converting text into numerical vectors that represent meaning. This allows systems to compare the semantic similarity between documents, questions, and sentences.

Text: "JWT is used for authentication"

Embedding: [0.23, -0.11, 0.78, ...]

- Embeddings help search by meaning instead of exact keywords.
- They are used in semantic search, recommendation, resume-job matching, duplicate detection, and RAG.
- Similar meanings produce vectors that are mathematically close to each other.

4.5 Vector Database

A vector database stores embeddings and helps search for semantically similar information. It is different from a traditional database because it searches by meaning, not only by exact values.

Traditional Database	Vector Database
Stores structured data such as rows and columns	Stores numerical embeddings/vectors
Searches exact values such as email or ID	Searches similar meaning using vector similarity
Used for users, orders, payments, products	Used for RAG, semantic search, document Q&A
Examples: MySQL, PostgreSQL, SQLite	Examples: FAISS, ChromaDB, Pinecone, Qdrant, Weaviate

4.6 RAG - Retrieval-Augmented Generation

RAG stands for Retrieval-Augmented Generation. It is a technique where relevant information is first retrieved from external documents or databases, and then the LLM uses that context to generate a more accurate answer.

Simple line: RAG = retrieve relevant data first, then generate the answer.

Phase	Description
Indexing Phase	Documents are extracted, split into chunks, converted into embeddings, and stored in a vector database.
Query Phase	User question is converted into embedding, similar chunks are retrieved, and the LLM generates an answer using that context.

- RAG is useful for PDF chatbots, company knowledge base assistants, documentation search, legal document assistants, and college notes Q&A.
- It reduces hallucination because answers are grounded in retrieved context.
- It allows the AI to use private or latest documents without retraining the model.

5. Introduction to Agentic AI

Agentic AI refers to AI systems that can work toward a goal by planning steps, using tools, remembering context, taking actions, observing results, and adjusting the next step. Unlike a simple Generative AI system that only produces content, an Agentic AI system tries to complete a task.

Simple definition: Agentic AI is goal-oriented AI that can plan and act using tools.

- It understands the user goal.
- It breaks the goal into steps.
- It selects and uses tools when needed.
- It observes tool results and decides the next action.
- It can ask for human approval before risky actions.

6. Components of Agentic AI

Component	Meaning	Example
Goal	The final target given by the user	Find top 5 real estate leads
Planning	Breaking a big goal into smaller steps	Collect data, filter, score, recommend
Tools	External services/functions used by the agent	Web search, database, email, calculator, API
Memory	Remembering useful context or preferences	User prefers simple explanations
Action	Actual step performed by the agent	Create draft, call API, search database
Observation	Result received after an action/tool call	Database returned 20 leads
Feedback	Checking whether output meets the goal	If results are weak, search again or refine filters

6.1 Tools in Agentic AI

Tools are external functions or services that allow an AI agent to do real work beyond text generation. A model is like the brain, while tools are like hands.

Tool	Use
Web Search	Find latest or external information
Calculator	Perform accurate calculations
Database Tool	Fetch, update, or store application data
Email Tool	Create drafts or send emails after approval
Calendar Tool	Create meetings or reminders
File Reader	Read PDFs, documents, or CSV files
API Tool	Call external services such as weather, payment, CRM, maps
Code Execution	Run code, test scripts, or analyze files

Safety rule: AI can draft or prepare actions, but risky actions should require human approval.

6.2 Memory in Agentic AI

Memory allows an agent to remember previous context, user preferences, past actions, or useful information. It helps the agent continue tasks instead of treating every request as new.

Memory Type	Meaning	Example
Short-term Memory	Current conversation context	User asks 'explain this' and AI knows 'this' means JWT
Long-term Memory	Saved useful information across sessions	User prefers simple English and interview-ready answers

6.3 Planning in Agentic AI

Planning means breaking a user goal into smaller steps and deciding the order of actions needed to complete the task. Planning helps Agentic AI handle complex tasks in an organized way.

Goal: Prepare a report on API Integration

Plan:

1. Define API integration
2. Explain REST APIs
3. Explain request and response
4. Add JSON handling
5. Add authentication methods
6. Add third-party API examples
7. Add interview Q&A

- Static planning follows a fixed plan.
- Dynamic planning adjusts the next step based on results.
- Planning plus tools allows the agent to complete real workflows.

7. Generative AI vs Agentic AI

Point	Generative AI	Agentic AI
Main Purpose	Create content	Complete goals or tasks
Input	Prompt	Goal or task
Output	Text, code, image, summary, document	Completed workflow, action result, report, decision
Planning	Limited or optional	Core feature
Tool Use	Optional	Very important
Memory	Optional	Often useful
Autonomy	Low to medium	Higher
Example	Write an email	Write email, create draft, ask approval
Main Risk	Hallucination or biased content	Wrong action, tool misuse, privacy issue

Most important line: Generative AI creates. Agentic AI acts toward a goal.

Agentic AI often uses Generative AI as its language and reasoning engine, but adds planning, memory, tools, actions, workflows, and safety controls on top of it.

8. Technologies and Frameworks Used

8.1 Technologies Used in Generative AI

Technology	Purpose
Python	Common programming language for AI, ML, APIs, and automation
LLMs	Core models for text generation, code generation, summarization, and Q&A
Hugging Face Transformers	Access and use pretrained NLP, vision, audio, and multimodal models
PyTorch / TensorFlow	Build, train, and fine-tune deep learning models
Embeddings	Convert meaning of text into vectors for similarity search
Vector Databases	Store embeddings for semantic search and RAG
RAG	Retrieve external knowledge and generate grounded answers
FastAPI	Expose AI features as backend APIs
Streamlit / React	Build user interfaces for AI demos and applications

8.2 Technologies Used in Agentic AI

Framework / Tool	Use in Agentic AI
LangChain	Build LLM apps with prompts, tools, memory, retrievers, and RAG pipelines
LangGraph	Build controlled, stateful, graph-based agent workflows
CrewAI	Create multi-agent teams where agents have roles and collaborate
OpenAI Agents SDK	Build agents with instructions, tools, handoffs, state, and guardrails
AutoGen	Build multi-agent conversational workflows
LlamaIndex	Connect LLMs with documents and data for RAG and knowledge assistants
APIs and Tools	Allow agents to call databases, email, web search, calendars, and external services
Guardrails	Add safety checks before risky actions
Monitoring Tools	Track errors, cost, latency, and agent behavior

Framework idea: Agentic AI frameworks help convert an LLM from a simple text generator into a task-completing system.

9. Real-world Use Cases

9.1 Generative AI Use Cases

Use Case	Description
Content Writing	Generate blogs, emails, captions, product descriptions, and reports
Code Generation	Generate code snippets, API routes, SQL queries, and debugging help
Documentation	Create technical notes, project reports, and user manuals
Summarization	Summarize PDFs, meetings, articles, research papers, and feedback
Education	Explain concepts, generate quizzes, and provide interview practice
Customer Support	Generate response drafts and answer common questions
Image/Design	Create posters, logos, product mockups, and illustrations

9.2 Agentic AI Use Cases

Use Case	Agentic Behavior
Sales/Lead Generation Agent	Collects leads, scores them, filters high-quality leads, and prepares outreach messages
Research Assistant	Searches sources, extracts points, compares information, and prepares a report
Coding Agent	Writes code, runs tests, observes errors, fixes bugs, and documents changes
Customer Support Agent	Checks order data, understands issue, creates ticket, and drafts response
Travel Planning Agent	Compares budget, hotel, transport, and itinerary options
Data Analysis Agent	Loads data, cleans it, generates charts, finds insights, and prepares summary
Personal Productivity Agent	Schedules tasks, creates reminders, drafts emails, and tracks progress

10. Risks, Limitations, and Safety Practices

10.1 Risks in Generative AI

- Hallucination: the model may produce confident but wrong information.
- Bias: generated output may reflect bias present in training data.
- Outdated information: model may not know recent facts unless connected to tools or updated data.
- Privacy issues: sensitive data should not be shared carelessly with AI systems.
- Over-dependence: outputs should be reviewed instead of accepted blindly.

10.2 Risks in Agentic AI

- Wrong tool use: the agent may call the wrong API or use wrong data.
- Unsafe actions: sending emails, deleting records, or making payments should need approval.
- Privacy leakage: agents with tools may access sensitive information.
- Cost overuse: repeated tool calls or long workflows may increase cost.
- Poor planning: wrong or incomplete plans can lead to weak results.

10.3 Safety Practices

- Use human-in-the-loop approval for risky actions.
- Apply guardrails to validate outputs and control tool permissions.
- Log agent actions for monitoring and debugging.
- Limit access to sensitive tools and databases.
- Verify important answers using reliable sources or retrieved documents.
- Avoid storing unnecessary personal or sensitive data in memory.

Important safety line: AI can prepare actions, but humans should approve risky real-world actions.

11. Example Architectures

11.1 Simple Generative AI Chatbot

User Interface: Streamlit or React

Backend: FastAPI

Model: LLM API or local model

Flow:

User question -> Backend prompt -> LLM response -> UI displays answer

11.2 RAG-based PDF Chatbot

Documents/PDFs

-> Text extraction

-> Chunking

-> Embedding generation

-> Vector database storage

-> User question

-> Retrieve similar chunks

-> LLM generates grounded answer

11.3 Agentic Lead Generation Assistant

Goal: Find high-quality real estate leads

Tools: Database, lead scoring logic, CRM API, email draft tool

Agent Flow:

1. Fetch lead data

2. Filter by location and budget

3. Score leads

4. Select top leads
5. Generate personalized outreach messages
6. Ask human approval before sending

12. Interview-ready Questions and Answers

Q: What is Generative AI?

A: Generative AI is a type of AI that creates new content such as text, code, images, audio, summaries, or documents based on user prompts.

Q: What is Agentic AI?

A: Agentic AI is an AI system that can understand a goal, plan steps, use tools, take actions, observe results, and adjust its behavior to complete a task.

Q: Difference between Generative AI and Agentic AI?

A: Generative AI creates content, while Agentic AI completes goals using planning, memory, tools, and actions.

Q: What is an LLM?

A: An LLM is a Large Language Model trained on large text data to understand prompts and generate human-like text, code, summaries, and answers.

Q: What is a prompt?

A: A prompt is the instruction or input given to an AI model. A good prompt includes task, context, format, tone, and constraints.

Q: What is a token?

A: A token is a small unit of text processed by a language model. Tokens affect context length, output length, and cost.

Q: What are embeddings?

A: Embeddings convert text meaning into numerical vectors so machines can compare similarity and perform semantic search.

Q: What is a vector database?

A: A vector database stores embeddings and helps find semantically similar information for RAG, document search, and AI memory.

Q: What is RAG?

A: RAG stands for Retrieval-Augmented Generation. It retrieves relevant external information first and then uses that context to generate accurate answers.

Q: Why are tools important in Agentic AI?

A: Tools allow agents to perform actions such as web search, database queries, calculations, email drafting, API calls, and file reading.

Q: What is memory in Agentic AI?

A: Memory allows an agent to remember previous context, user preferences, or past actions to provide personalized and continuous assistance.

Q: What is planning in Agentic AI?

A: Planning means breaking a big goal into smaller steps and deciding the order of actions needed to complete the task.

Q: Name some Agentic AI frameworks.

A: LangChain, LangGraph, CrewAI, OpenAI Agents SDK, AutoGen, and LlamaIndex are commonly used for LLM applications, workflows, and agent systems.

Q: What are guardrails?

A: Guardrails are safety checks that control AI outputs and actions, especially when agents use tools or perform risky tasks.

13. Final Summary

Generative AI and Agentic AI are closely related but serve different purposes. Generative AI focuses on creating new content such as text, code, summaries, images, and documents. Agentic AI goes beyond generation by working toward goals through planning, memory, tools, actions, and feedback.

In practical AI applications, LLMs act as the language and reasoning engine, prompts guide the output, tokens are the processing units, embeddings represent meaning as vectors, vector databases enable semantic search, and RAG connects LLMs with external knowledge. Agentic AI adds tools, memory, planning, workflows, and safety controls to complete real tasks.

Final interview line: Generative AI creates content, while Agentic AI uses Generative AI plus planning, tools, memory, and actions to complete goals.